

Q2. Typesetting

The so-called word refers to one or more English letters (letters).

If you want to beautify the output and print, you need to adjust the gap between letters and words.

The unit of this gap is called dots. Given the total length of each line of text (in points), the gap between letters in a word should be the largest, and the gap should be less than or equal to $\frac{1}{3}$ of the gap between words, but not Less than 3; and the minimum number of points for the gap between words is 10 (there is no upper limit for the number of gap points between words).

If there are gaps between multiple words, the gaps in the back cannot exceed the gaps in the front, and it should be as even as possible starting from the head; the gaps of all letters in the same row must be the same to achieve aesthetics. If the total length given is not enough, print with the smallest gap, but it cannot violate the above principles.

The width of all letters that will appear is defined as follows:

A: 17 dots; B: 18 dots; C: 16 dots; W: 21 dots; X: 18 dots;

The beginning and end of the entire line cannot contain a gap, and the last end mark is a continuous line of 0 and a line of YZYZ; the end mark does not need to be printed.

Input:

Each input consists of 2 lines, and you can enter multiple inputs continuously, and the last line ends with a line of 0 and a line of "YZYZ".

The first line of each input is the total width, and the second line is the text to be processed and printed. E.g:

100

AA BB

0

YZYZ

The first 2 lines indicate that AA BB should be printed in a width of 100 dots, and the last 2 lines indicate the end.

Output:

Write the gap between letters in (), for example, print AA BB in a width of 100 dots, it will output:

A(6)A(18)B(6)B

Sample input:

100

AA BB

90

ABB W

0

YZYZ

Sample output:

A(6)A(18)B(6)B

A(3)B(3)B(10)W